HW 2

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> **Townsend 2.5 (5 pts)**
> **Problem 2.5** What is the matrix representation of $\hat{J}_z$ using the states $|+\mathbf{y}\rangle$ and $|-\mathbf{y}\rangle$ as a basis? Use this representation to evaluate the expectation value of S_z for a collection of particles each in the state $|-\mathbf{y}\rangle$.

The state $|y\rangle$ has the matrix representation in the z basis

$$\begin{aligned}
|\vec{y}\rangle &= \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix} \\
|-\vec{y}\rangle &= \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix}
\end{aligned}$$

Or in the $|\vec{y}\rangle$ basis, just the identities

$$\begin{aligned}
|+y\rangle &= \begin{pmatrix}1\\ 0\end{pmatrix} \\
|-y\rangle &= \begin{pmatrix}0\\ 1\end{pmatrix}
\end{aligned}$$

The rotator $\hat{J}_z$ in the $|\vec{z}\rangle$ basis is

$$\begin{pmatrix}\frac{\hbar}{2} & 0\\ 0 & -\frac{\hbar}{2}\end{pmatrix}\begin{pmatrix}|+z\rangle\\ |-z\rangle\end{pmatrix}$$

We can project to the $|\vec{y}\rangle$ basis by using the rotation matrix

$$\begin{aligned}
\mathbb{S} &= \begin{pmatrix}\langle +z|+y\rangle & \langle +z|-y\rangle\\ \langle -z|+y\rangle & \langle -z|-y\rangle\end{pmatrix} \\
&= \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ i & -i\end{bmatrix}
\end{aligned}$$

We can now compute

$$\begin{aligned}
J_z &\xrightarrow{S_y\text{ basis}} \frac{1}{2}\begin{bmatrix}1 & -i\\ 1 & i\end{bmatrix}\frac{\hbar}{2}\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}1 & 1\\ i & -i\end{bmatrix} \\
&= \frac{\hbar}{4}\begin{bmatrix}1 & -i\\ 1 & i\end{bmatrix}\begin{bmatrix}1 & 1\\ -i & i\end{bmatrix} \\
&= \frac{\hbar}{4}\begin{bmatrix}0 & 2\\ 2 & 0\end{bmatrix} \\
&= \frac{\hbar}{2}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}
\end{aligned}$$

We can now evaluate $\langle S_z\rangle$ for particles in the $|-y\rangle$ state as

$$\begin{aligned}
\langle -y|J_z|-y\rangle &= [0\ \ 1]\frac{\hbar}{2}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}0\\ 1\end{bmatrix} \\
&= [0\ \ 1]\frac{\hbar}{2}\begin{bmatrix}1\\ 0\end{bmatrix} \\
\langle S_z\rangle &= 0
\end{aligned}$$

Townsend 2.6 (5 pts)
Evaluate $\hat{R}(\theta\mathbf{j})|+\mathbf{z}\rangle$, where $\hat{R}(\theta\mathbf{j}) = e^{-i\hat{J}_y\theta/\hbar}$ is the operator that rotates kets counterclockwise by angle θ about the y axis. Show that $\hat{R}\left(\frac{\pi}{2}\mathbf{j}\right)|+\mathbf{z}\rangle = |+\mathbf{x}\rangle$. *Suggestion*: Express the ket $|+\mathbf{z}\rangle$ as a superposition of the kets $|+\mathbf{y}\rangle$ and $|-\mathbf{y}\rangle$ and take advantage of the fact that $\hat{J}_y|\pm\mathbf{y}\rangle = (\pm\hbar/2)|\pm\mathbf{y}\rangle$; then switch back to the $|+\mathbf{z}\rangle$ and $|-\mathbf{z}\rangle$ basis.

We can take the vector $|+z\rangle$ to the $|\vec{y}\rangle$ basis with the $\mathbb{S}$ rotation matrix

$R\left(\frac{\pi}{2}j\right) = e^{-i\hat{J}_y\frac{\pi}{2\hbar}}$

$$\begin{bmatrix} \langle +y|+z\rangle & \langle +y|-z\rangle \\ \langle -y|+z\rangle & \langle -y|-z\rangle \end{bmatrix}$$

$$\mathbb{S} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -i \\ 1 & i \end{bmatrix}$$

Note that $\mathbb{S}^{-1} = \mathbb{S}^{\dagger}$, so

$$\mathbb{S}^{-1} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix}$$

We can now write the matrix from $|+z\rangle$ in the z basis to the $|y\rangle$ basis.

$$|+z\rangle_y = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -i \\ 1 & i \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$
$$|+z\rangle_y = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

We can also take the vector $|+x\rangle$ to the $|\vec{y}\rangle$ basis

$$|+x\rangle_y = \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -i \\ 1 & i \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$
$$= \frac{1}{2}\begin{bmatrix} 1-i \\ 1+i \end{bmatrix}$$

We can now see how the $\hat{R}\left(\frac{\pi}{2}j\right)$ rotator acts, since this is now expressed in the Eigenbasis.
We know that $\hat{J}_y|\pm y\rangle = \pm\frac{\hbar}{2}|\pm y\rangle$, which makes the matrix representation for the operator in the S_y basis:

$$\hat{J}_y \xrightarrow{S_y \text{ basis}} \begin{pmatrix} \langle +y|\hat{J}_y|+y\rangle & \langle -y|\hat{J}_y|+y\rangle \\ \langle +y|\hat{J}_y|-y\rangle & \langle -y|\hat{J}_y|-y\rangle \end{pmatrix}$$
$$= \begin{pmatrix} \frac{\hbar}{2} & 0 \\ 0 & -\frac{\hbar}{2} \end{pmatrix}$$

$R\left(\frac{\pi}{2}j\right) = e^{-i\hat{J}_y\frac{\pi}{2\hbar}}$, so this becomes

$$\hat{R}(\phi\hat{J}) = \begin{pmatrix} e^{\frac{-i\phi}{2}} & 0 \\ 0 & e^{\frac{i\phi}{2}} \end{pmatrix}$$
$$\implies R\left(\frac{\pi}{2}\hat{J}\right) = \begin{pmatrix} e^{\frac{-i\pi}{4}} & 0 \\ 0 & e^{\frac{i\pi}{4}} \end{pmatrix}$$

We can now finally compute $\hat{R}\left(\frac{\pi}{2}\hat{J}\right)|+z\rangle$ nicely in the $|y\rangle$ basis

$$
\begin{aligned}
|+z\rangle_y &= \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}\\
\hat{R}\left(\frac{\pi}{2}\hat{J}\right)|+z\rangle_y &= \frac{1}{2}\begin{pmatrix}e^{\frac{-i\pi}{4}} & 0\\ 0 & e^{\frac{i\pi}{4}}\end{pmatrix}\begin{bmatrix}1\\1\end{bmatrix}\\
&= \frac{1}{2}\begin{bmatrix}e^{\frac{-i\pi}{4}}\\ e^{\frac{i\pi}{4}}\end{bmatrix}\\
&= \frac{1}{2}\sqrt{\frac{1}{2}}\begin{bmatrix}1+i\\1-i\end{bmatrix}
\end{aligned}
$$

We can see that this is the same as out $|+x\rangle_y$, so rotating $|+z\rangle$ by $\frac{\pi}{2}$ around the $|\vec{y}\rangle$ axis yields $|x\rangle$, as geometry demands.

Townsend 2.12 (5 pts)
A photon polarization state for a photon propagating in the z direction is given by

$$|\psi\rangle = \sqrt{\frac{2}{3}}|x\rangle + \frac{i}{\sqrt{3}}|y\rangle$$

(a) What is the probability that a photon in this state will pass through an ideal polarizer with its transmission axis oriented in the y direction?

(b) What is the probability that a photon in this state will pass through an ideal polarizer with its transmission axis y' making an angle ϕ with the y axis?

(c) A beam carrying N photons per second, each in the state $|\psi\rangle$, is totally absorbed by a black disk with its normal to the surface in the z direction. How large is the torque exerted on the disk? In which direction does the disk rotate? *Reminder*: The photon states $|R\rangle$ and $|L\rangle$ each carry a unit $\hbar$ of angular momentum parallel and antiparallel, respectively, to the direction of propagation of the photons.

(d) How would the result for each of these questions differ if the polarization state were

$$|\psi'\rangle = \sqrt{\frac{2}{3}}|x\rangle + \frac{1}{\sqrt{3}}|y\rangle$$

that is, the "i" in the state $|\psi\rangle$ is absent?

A

$$\text{Prob (y)} = \left|\left(\frac{i}{\sqrt{3}}\right)^2\right|$$

more formally: $|\langle +y|\psi\rangle|^2$
Which is $\frac{1}{3}$.

B

We can write the general state of the photon in terms of the new axis of the polarizer

$$
\begin{aligned}
|x'\rangle &= \cos\phi|x\rangle + \sin\phi|y\rangle\\
|y'\rangle &= -\sin\phi|x\rangle + \cos\phi|y\rangle
\end{aligned}
$$

This question reduces to the statement

$$
\begin{aligned}
|\langle y'|\psi\rangle|^2 &= \left|\left((-\sin\phi\langle x| + \cos\phi\langle y|)\left(\sqrt{\frac{2}{3}}|x\rangle + \frac{i}{\sqrt{3}}|y\rangle\right)\right)\right|^2\\
&= \left|\left(-\sin\phi\sqrt{\frac{2}{3}} + \cos\phi\frac{i}{\sqrt{3}}\right)\right|^2\\
&\sin^2\phi\frac{2}{3} + \cos^2\phi\frac{1}{3}
\end{aligned}
$$

C

The total angular momentum imparted per second will be roughly
$N_L\hbar - N_R\hbar$

We can find N_L and N_R.
$N_L = N - N_R$

$$
\begin{aligned}
|R\rangle &= \frac{1}{\sqrt{2}}(|x\rangle + i|y\rangle) \\
|L\rangle &= \frac{1}{\sqrt{2}}(|x\rangle - i|y\rangle)
\end{aligned}
$$

We know that

$$|\psi\rangle = \sqrt{\frac{2}{3}}|x\rangle + \frac{i}{\sqrt{3}}|y\rangle$$

We can thus find

$$
\begin{aligned}
|\langle R|\psi\rangle|^2 &= \left(\frac{1}{\sqrt{2}}(\langle x| - i\langle y|)\left(\sqrt{\frac{2}{3}}|x\rangle + \frac{i}{\sqrt{3}}|y\rangle\right)\right)^2 \\
&= \left(\frac{1}{\sqrt{2}}\sqrt{\frac{2}{3}} + \frac{1}{\sqrt{6}}\right)^2 \\
&= \left(\frac{1}{\sqrt{6}}(\sqrt{2}+1)\right)^2 \\
&= \frac{1}{6}(3 + 2\sqrt{2})
\end{aligned}
$$

This gives that there are $\sim 0.97N$ right circularly polarized photons, and $\sim 0.03N$ Left circularly polarized.

The momentum change per second would thus be $(0.97 - 0.03)N\hbar = 0.94N\hbar$

D

Every calculation here uses the complex magnitude, so any i factor cancels out with a $-i$ to be the same. We don't have any terms which would change with the overall phase difference between $1 \text{ and } i$. A,B, and C would have identical resulting probabilities.

Townsend 2.13 (5 pts)
A system of N ideal linear polarizers is arranged in sequence, as shown in Fig. 2.13. The transmission axis of the first polarizer makes an angle of ϕ/N with the y axis. The transmission axis of every other polarizer makes an angle of ϕ/N with respect to the axis of the preceding one. Thus, the transmission axis of the final polarizer makes an angle ϕ with the y axis. A beam of y-polarized photons is incident on the first polarizer.

(a) What is the probability that an incident photon is transmitted by the array?

(b) Evaluate the probability of transmission in the limit of large N.

(c) Consider the special case with the angle $\phi = 90°$. Explain why your result is not in conflict with the fact that $\langle x|y\rangle = 0$.

result is counterintuitive, no sense classically.
The probability of passing through the first polarizer from a state $|+y\rangle$ would be the same as the probability of passing from the first through the second, since we assume that all the offset angles are the exact same. Thus the chance of passing through N polarizers is

$$\left(\langle y'| + y\rangle\right)^N$$

We can evaluate this initially in the $|y\rangle$ basis, and this assumes that each successive probability is in its own $y \underbrace{n'}_{\text{some number}}$ basis - i.e. $\langle y''|y'\rangle$, $\langle y'''|y''\rangle$, where each has the same probability of occurrence as the previous event.

$$\langle y'| = -\sin\theta\langle x| + \cos\theta\langle y|$$

This means that

$$\langle y'|y\rangle = \cos\frac{\phi}{N}$$

So the chance of going through the series of polarizers is

$$\cos^N\frac{\phi}{N}$$

For $\phi = \frac{\pi}{2}$,

$$\begin{aligned}\text{Probability of passing} \\ &= \lim_{n\to\infty}\cos^N\frac{\pi}{2N}\end{aligned}$$

To sketch the argument in words:
$\lim_{N\to\infty}\frac{\pi}{2N} = 0$, and $\cos(0) = 1$
So as $N \to \infty$, the probability of transmission through each successive filter approaches 1. Thus, the probability through all filters approaches 1.

We can prove this by Taylor expanding and using the binomial theorem.

$$\cos^N\frac{\pi}{2} = \left(\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\pi}{2}\right)^{2k}}{2k!}\right)^N$$

If we only keep up to second order terms of $\cos$, this becomes

$$\cos^N(x) \approx \left(1 - \frac{x^2}{2} + \mathscr{O}(x^3)\right)^N$$

We can simplify this with the binomial expansion

$$\lim_{n\to\infty}\cos^n\frac{\pi}{2n} \approx \lim_{n\to\infty}\left(1 - \frac{\pi^2}{4n^2}\right)^N$$

$$\begin{aligned}&= \lim_{n\to\infty} 1 + \cancelto{0}{\frac{\pi^2}{4n^2}} + \cancelto{0}{\mathscr{O}\left(\left(\frac{\pi^2}{4n^2}\right)^2\right)} \\ &= 1\end{aligned}$$

Thus, there is a 100% chance that a photon will pass through infinite polarizers that are offset by infinitely small angles all the way down to $\frac{\pi}{2}$.

$\langle x|y\rangle = 0$ means that a particle in state y has no chance of being in state x. However, this isn't going between those two - it is between adjacent states of polarization $\langle y'|y\rangle$, for which there is a finite chance. As $\theta \to 0$, $\cos\theta = 1$.

Townsend 2.21 (5 pts)
Linearly polarized light of wavelength 5890 Å is incident normally on a birefringent crystal that has its optic axis parallel to the face of the crystal, along the x axis. If the incident light is polarized at an angle of 45° to the x and y axes, what is the probability that the photons exiting a crystal of thickness 100.0 microns will be right-circularly polarized? The index of refraction for light of this wavelength polarized along y (perpendicular to the optic axis) is 1.66 and the index of refraction for light polarized along x (parallel to the optic axis) is 1.49.

referencing pg 63 of Townsend

The initial state is $|\psi\rangle = \frac{1}{\sqrt{2}}(|x\rangle + |y\rangle)$

Light parallel to the x axis will pick up a phase $n_x \frac{\omega}{c} z$ traveling a distance z.

in the y axis, $n_{y \frac{\omega}{c}} z$.

At 45 degrees, there is a phase difference $\left[(n_x - n_y)\frac{\omega}{c}\right] z$ between x and y.

If the phase difference is $\frac{\pi}{2}$, then circularly polarized light will be emitted.

$\lambda = 5890\dot{A}$.

$\omega = 2\pi\nu = 2\pi\frac{c}{\lambda}$

$\frac{\omega}{c} = \frac{2\pi}{\lambda}$

Thus we can simplify

$$
\begin{aligned}
\theta &= \frac{[n_y - n_x]\omega}{c} z \\
&= [-0.17]\frac{2\pi}{5890\dot{A}} 100\mu m \\
&= -0.17 \cdot 2\pi \cdot 58.9 \\
\theta &= 62.9135
\end{aligned}
$$

This means that $|\psi'\rangle = \frac{1}{\sqrt{2}}(|x\rangle + e^{62.9135}|y\rangle)$

We know the state

$$
|R\rangle = \frac{1}{\sqrt{2}}(|x\rangle + i|y\rangle)
$$

So we can find

$$
\begin{aligned}
\langle R| &= \frac{1}{\sqrt{2}}(\langle x| - \langle y|) \\
\langle R|\psi'\rangle^2 &= \left|\frac{1}{2} - \frac{1}{2} i e^{-63i}\right|^2 \\
&= \frac{1}{4}\left|1 - ie^{-63i}\right|^2 \\
&= \frac{1}{4}\left|1 + ie^{63i} - ie^{63i}\right| \\
&= \frac{1}{4}|1 + \sin 63| \text{ (radians)} \\
&\approx 30\% \text{ chance}
\end{aligned}
$$